

Praktikum 14 (Versi Sederhana)

Tujuan

Menginstal dan menjalankan Intrusion Detection System (IDS) menggunakan Snort untuk mendeteksi aktivitas jaringan sederhana. Sesuai konsep pada jurnal, Snort digunakan untuk mendeteksi aktivitas mencurigakan berdasarkan rules yang dibuat.

Langkah 1

Jalankan Ubuntu.

```
4:23:53 +0000. DataSource DataSourceNone. Up 23.55 seconds
webserver-p6 login: falah
Password:
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 5.15.0-185-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:   https://landscape.canonical.com
 * Support:      https://ubuntu.com/pro

System information as of Mon Jul 20 02:19:36 PM UTC 2026

System load:          0.85
Usage of /:            46.7% of 11.21GB
Memory usage:         11%
Swap usage:           0%
Processes:            118
Users logged in:      0
IPv4 address for enp0s3: 10.0.2.15
IPv6 address for enp0s3: fd00::a00:27ff:fe74:1d59

Expanded Security Maintenance for Applications is not enabled.

63 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The list of available updates is more than a week old.
To check for new updates run: sudo apt update

Last login: Sat Jul 4 23:39:36 UTC 2026 on tty1
falah@webserver-p6:~$
falah@webserver-p6:~$ _
```

Desktop Ubuntu berhasil masuk.

Langkah 2

Update Ubuntu

Buka Terminal

sudo apt update

Setelah selesai

`sudo apt upgrade -y`

Proses update.

```
* Support:      https://ubuntu.com/pro

System information as of Mon Jul 20 02:19:36 PM UTC 2026

System load:      0.85
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Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The list of available updates is more than a week old.
To check for new updates run: sudo apt update

Last login: Sat Jul  4 23:39:36 UTC 2026 on tty1
falah@webserver-p6:~$
falah@webserver-p6:~$ sudo apt update
[sudo] password for falah:
Get:1 http://security.ubuntu.com/ubuntu jammy-security InRelease [129 kB]
Hit:2 http://id.archive.ubuntu.com/ubuntu jammy InRelease
Get:3 http://id.archive.ubuntu.com/ubuntu jammy-updates InRelease [128 kB]
Get:4 http://id.archive.ubuntu.com/ubuntu jammy-backports InRelease [127 kB]
Get:5 http://security.ubuntu.com/ubuntu jammy-security/main amd64 Packages [3,361 kB]
Get:6 http://id.archive.ubuntu.com/ubuntu jammy-updates/main amd64 Packages [3,646 kB]
39% [6 Packages 1,428 kB/3,646 kB 39%] [Waiting for headers] 149 kB/s 1min 31s
```

Langkah 3

Install Snort

`sudo apt install snort -y`

Kalau muncul pertanyaan Network Address

Misalnya

192.168.1.0/24

Kalau bingung kirim screenshot ke saya.

Package configuration

Configuring snort

Please use the CIDR form - for example, 192.168.1.0/24 for a block of 256 addresses or 192.168.1.42/32 for just one. Multiple values should be comma-separated (without spaces).

You can leave this value empty and configure HOME_NET in /etc/snort/snort.conf instead. This is useful if you are using Snort in a system which frequently changes network and does not have a static IP address assigned.

Please note that if Snort is configured to use multiple interfaces, it will use this value as the HOME_NET definition for all of them.

Address range for the local network:

192.168.0.0/16

<OK>

```
Setting up libfile-listing-perl (6.14-1) ...
Setting up libnet-http-perl (6.22-1) ...
Setting up liblua5.1-2:amd64 (2.1.0~beta3+dfsg-6ubuntu0.1) ...
Setting up libdaq2 (2.0.7-5) ...
Setting up libwww-robotrules-perl (6.02-1) ...
Setting up libhtml-parser-perl:amd64 (3.76-1build2) ...
Setting up snort-common-libraries (2.9.15.1-6build1) ...
Setting up libio-socket-ssl-perl (2.074-2) ...
Setting up libhttp-message-perl (6.36-1) ...
Setting up libhtml-form-perl (6.07-1) ...
Setting up libhttp-negotiate-perl (6.01-1) ...
Setting up snort (2.9.15.1-6build1) ...
Snort configuration: interface default not set, using 'enp0s3'
Setting up libhttp-cookies-perl (6.10-1) ...
Setting up libhtml-tree-perl (5.07-2) ...
Setting up libhtml-format-perl (2.12-1.1) ...
Setting up libnet-smtp-ssl-perl (1.04-1) ...
Setting up libmailtools-perl (2.21-1) ...
Setting up libhttp-daemon-perl (6.13-1ubuntu0.2) ...
Setting up liblwp-protocol-https-perl (6.10-1) ...
Setting up libwww-perl (6.61-1ubuntu0.1) ...
Setting up oinkmaster (2.0-4.1) ...
Processing triggers for libc-bin (2.35-0ubuntu3.13) ...
Processing triggers for man-db (2.10.2-1) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
falah@webserver-p6:~$ _
```

Install Snort berhasil.

Langkah 4

Cek apakah Snort sudah terinstall

snort -V

Kalau berhasil muncul

Version 2.x.x

```
0 upgraded, 0 newly installed, 0 to remove and 92 not upgraded.
falah@webserver-p6:~$ snort-v
snort-v: command not found
falah@webserver-p6:~$ which snort
/usr/sbin/snort
falah@webserver-p6:~$ /usr/sbin/snort -v
Running in packet dump mode

==== Initializing Snort ====
Initializing Output Plugins!
pcap DAQ configured to passive.
Acquiring network traffic from "enp0s3".
ERROR: Can't start DAQ (-1) - socket: Operation not permitted!
Fatal Error, Quitting..
falah@webserver-p6:~$ sudo snort -v
Running in packet dump mode

==== Initializing Snort ====
Initializing Output Plugins!
pcap DAQ configured to passive.
Acquiring network traffic from "enp0s3".
Decoding Ethernet

==== Initialization Complete ====

o''')~ -> Snort! <*-
      ~ Version 2.9.15.1 GRE (Build 15125)
      ~ By Martin Roesch & The Snort Team: http://www.snort.org/contact#team
      ~ Copyright (C) 2014-2019 Cisco and/or its affiliates. All rights reserved.
      ~ Copyright (C) 1998-2013 Sourcefire, Inc., et al.
      ~ Using libpcap version 1.10.1 (with TPACKET_V3)
      ~ Using PCRE version: 8.39 2016-06-14
      ~ Using ZLIB version: 1.2.11

Commencing packet processing (pid=2494)
```

Hasil snort -V

Langkah 5

Masuk folder rules

cd /etc/snort/rules

Lalu

sudo nano local.rules

Tambahkan satu rule sederhana

alert icmp any any -> any any (msg:"PING TERDETEKSI"; sid:1000001; rev:1;)

Tekan

Enter

CTRL+X

```
# -----
# LOCAL RULES
# -----
# This file intentionally does not come with signatures.  Put your local
# additions here.
alert icmp any any -> any any (msg:"PING TERDETEKSI" sid:1000001; rev:1;

falah@webserver-p6:~$ sudo grep local.rules /etc/snort/snort.conf
include $RULE_PATH/local.rules
falah@webserver-p6:~$
```

Isi file local.rules

Langkah 6

Jalankan Snort

Cari nama interface jaringan:

ip a

```
falah@webserver-p6:~$ sudo grep local.rules /etc/snort/snort.conf
include $RULE_PATH/local.rules
falah@webserver-p6:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:74:1d:59 brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 metric 100 brd 10.0.2.255 scope global dynamic enp0s3
        valid_lft 83257sec preferred_lft 83257sec
    inet6 fd00::a00:27ff:fe74:1d59/64 scope global dynamic mngtmpaddr noprefixroute
        valid_lft 86063sec preferred_lft 14063sec
    inet6 fe80::a00:27ff:fe74:1d59/64 scope link
        valid_lft forever preferred_lft forever
falah@webserver-p6:~$
```

Biasanya namanya seperti:

- enp0s3
- enp0s8

Kalau sudah tahu, jalankan:

`sudo snort -A console -q -c /etc/snort/snort.conf -i enp0s3`

Ganti enp0s3 jika nama interfacenya berbeda.

```
falah@webserver-p6:~$ cat /etc/snort/rules/local.rules
# $Id: local.rules,v 1.11 2004/07/23 20:15:44 bmc Exp $
# -----
# LOCAL RULES
# -----
# This file intentionally does not come with signatures.  Put your local
# additions here.
alert icmp any any -> any any (msg:"PING TERDETEKSI"; sid:1000001; rev:1;)
falah@webserver-p6:~$ sudo snort -A console -q -c /etc/snort/snort.conf -i enp0s3
07/20-23:21:05.516509  [*] [1:1000001:1] PING TERDETEKSI [*] [Priority: 0] {IPV6-ICMP} fe80::2 ->
ff02::1
07/20-23:21:05.534630  [*] [1:1000001:1] PING TERDETEKSI [*] [Priority: 0] {IPV6-ICMP} fe80::a00:2
7ff:fe74:1d59 -> ff02::16
07/20-23:21:06.511446  [*] [1:1000001:1] PING TERDETEKSI [*] [Priority: 0] {IPV6-ICMP} fe80::a00:2
7ff:fe74:1d59 -> ff02::16
-
```

Snort sedang berjalan.

Langkah 7

Uji Deteksi

Dari **Windows (host)** buka Command Prompt:

Cari IP Ubuntu:

`ip a`

```
falah@webserver-p6:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:74:1d:59 brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 metric 100 brd 10.0.2.255 scope global dynamic enp0s3
        valid_lft 82677sec preferred_lft 82677sec
    inet6 fd00::a00:27ff:fe74:1d59/64 scope global dynamic mngtmpaddr noprefixroute
        valid_lft 86252sec preferred_lft 14252sec
    inet6 fe80::a00:27ff:fe74:1d59/64 scope link
        valid_lft forever preferred_lft forever
falah@webserver-p6:~$
```

Misalnya IP Ubuntu

inet 10.0.2.15/24

Di Windows jalankan

ping 10.0.2.15/24

Kalau rule benar,

di Ubuntu akan muncul

PING TERDETEKSI

Itu berarti IDS bekerja.